



A linear transportation L^p distance for pattern recognition

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ABSTRACT

The transportation L^p distance, denoted TL^p , has been proposed as a generalisation of Wasserstein W^p distances motivated by the property that it can be applied directly to colour or multi-channelled images, as well as multivariate time-series without normalisation or mass constraints. Both TL^p and W^p assign a cost based on the transport distance (i.e. the “Lagrangian” model), the key difference between the distances is that TL^p interprets the signal as a function whilst W^p interprets the signal as a measure. Both distances are powerful tools in modelling data with spatial or temporal perturbations. However, their computational cost can make them infeasible to apply to even moderate pattern recognition tasks. The linear Wasserstein distance was proposed as a method for projecting signals into a Euclidean space where the Euclidean distance is approximately the Wasserstein distance (more formally, this is a projection on to the tangent manifold). We propose linear versions of the TL^p distance (LTL^p) and we show significant improvement over the linear W^p distance on signal processing tasks, whilst being several orders of magnitude faster to compute than the TL^p distance.

1. Introduction

Optimal transport has gained in recent popularity because of its ability to model diverse data distributions in the signal and image processing fields [1]. Transportation-based methods have been successfully applied to image analysis, including medical images [2] and facial recognition [3], as well as cosmology [4,5] and voice recognition. Machine learning and Bayesian statistics have also benefited from transport-based approaches [6–11].

The popularity of optimal transport is, in part, due to the rise in the number of problems in the experimental and social sciences in which techniques are required to compare signal perturbations across spatial or temporal domains. For example, optimal transport based methods for image registration and warping [12] and image morphing [13] have existed for many years. Transportation techniques provide non-linear methods that jointly model locations and intensities, making transportation based approaches a powerful tool in many problems.

Optimal transport methods, in particular Wasserstein W^p distances, are grounded in a wealth of mathematical theory. Excellent introductions to the advanced mathematical theory of optimal transport are presented in [14,15], whilst [16] presents the theory with a more applied

perspective. Many technical aspects of optimal transport have been explored, including geometric properties [17] and links to evolutionary PDEs [18].

There is much interest in developing efficient methods to compute optimal transport distances and maps. For discrete measures, the optimal transport problem can be solved using linear programming approaches [16]. Since solving a linear programme can be costly, [19] proposed a multi-scale linear programme for efficient computations. Other approaches include flow minimisation techniques [12,18,20,21] gradient descent approaches [22] and multi-scale methods [23,24], amongst others [25–29]. More recently, Cuturi proposed entropy-regularisation based approaches to compute approximations of the optimal transport problem [30]. These methods have been explored in-depth with many extensions [8,31–37]. In order to efficiently compute pairwise optimal transport distances on a large data set, a framework called linear optimal transport was proposed in [38,39] for the W^p distance. In [38] the linear transportation distance was applied to classification tasks for medical, facial and galaxy images, [39] applied the distance to classification problems in medical, facial, galaxies and

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bird images, [40] used the framework to generate new images, and [41] used the distance to classify jets in collider data.

Higher-order transportation methods have been proposed in the mathematical analysis literature [42–44]. In [42] García Trillos and Slepčev proposed the transportation L^p (TL^p) distance to define discrete-to-continuum convergence of variational problems on point clouds. This was further extended in [43] to a transportation $W^{k,p}$ distance (where the notation relates to Sobolev spaces). Further developments and applications include [44] where the authors use the TL^p distance to define a (almost everywhere) differentiable misfit function. Using a linear sum assignment problem algorithm the gradient of the misfit function can be computed and the authors use this framework on Seismic imaging data. These transportation distances have several key advantages over the optimal transport distance; these include [43]:

1. They have no need for mass normalisation.
2. They are not restricted to non-negative measures.
3. They more accurately model translations compared to L^p .
4. They can compare signals with different discretisations.

Additionally, the transportation $W^{k,p}$ distance can include information on the derivative of the signal. The transportation distances TL^p and $W^{k,p}$ were proposed in [43] to tackle problems in signal analysis; for example, they were able to apply transportation methods to colour images. They experimentally show that the TL^p distance outperforms both L^p distances and W^p distances in classification tasks. However, these higher-order transportation distances require the computation of an optimal transport map (on a higher dimensional space) and this thwarts its application to larger scale pattern recognition tasks, where pairwise distances are needed.

Main contributions. In this paper, we propose a linear approximation of the TL^p distance, which is orders of magnitude faster to compute than the full TL^p distance first proposed in [42], whilst retaining some of its favourable properties. Our proposed method can be seen as an extension of the linear Wasserstein framework (LW^p) [38] to higher-order transportation distances. The linearisation is essentially a projection onto the tangent manifold at a given reference point. The geodesic distance (in this case corresponding to the Wasserstein distance) is approximated by the Euclidean distance in the tangent space. Suppose we wish to compare N signals/images, then application of optimal transport methods would require computation of $N(N-1)/2$ distances. In the linear optimal transport framework only N distances need to be computed. From here signals/images are then embedded into Euclidean space allowing linear statistical methods to be applied, whilst preserving much of the geometry of the original optimal transport space. In [38] the optimal transport distance of choice is the W^p distance, here we will choose the TL^p distance. This choice allows a more general set of unnormalised, not necessarily non-negative, multi-channel signals to be modelled, which is not possible in the linear W^p framework.

Our method produces a representation of the signal where the TL^p distance between the original signals is approximately equal to the Euclidean distance between the representations. Learning representations, particularly of images, is not restricted to optimal transport methods. In normed spaces one can consider lower dimensional projections that minimise error (the difference between the projection and the original signal). Among the most popular methods in Euclidean spaces is principal component analysis (PCA), and the same ideas motivate feature extraction in more general topologies, for example for the nuclear norm or $L^{2,1}$ -norms [45] and deep learning approaches [46]. The optimal transport topologies we use here are not induced by a norm but the motivation is similar: to derive a representation (or feature extraction) that preserves the distance. However, a key difference is that we are not looking for low-dimensional representations (although after linearisation our signals one can use traditional dimensionality reduction methods, such as PCA). More generally, autoencoders (see for example [47] for one of the most popular architectures) learn

representations, but typically without trying to preserve distances from the original space to the learned feature space.

Paper outline. This manuscript begins by reviewing optimal transport and the W^p and TL^p distances. The LW^p framework is reviewed in Section 2.3 and we propose our extension to TL^p in Section 2.4. We then give an overview on interpolation in the TL^p space. A background on numerical methods, more precisely how existing methods for W^p can be adapted to TL^p is included in the appendix. In Section 3 we apply our method to classification problems in Australian sign language (Section 3.1), breast cancer histopathology (Section 3.2) and financial time series (Section 3.3). We show that linear TL^p (LTL^p) outperforms LW^p and that it has similar performance to TL^p , but is several orders of magnitude faster. Other applications to synthetic data sets and cell morphometry are given in the appendix.

2. Methods

2.1. Optimal transport and the Wasserstein distance

To fix notation, we review the modern Monge–Kantorovich formulation of optimal transport and we refer to the excellent monographs [14–16,48] for a thorough exposition. Let μ and ν be probability measures on measure spaces X and Y respectively, i.e. $\mu, \nu \in \mathcal{P}(X)$. Further, let $\mathcal{B}(X)$ denote the Borel σ -algebra on X . We define the *pushforward* of a measure $\mu \in \mathcal{P}(X)$ by a function $h : X \rightarrow Z$ by $h_*\mu(A) := \mu(h^{-1}(A))$ for all $A \in \mathcal{B}(Z)$. The inverse of h is understood as being in the set theoretic sense, i.e. $h^{-1}(A) := \{x : h(x) \in A\}$. We denote by $\Pi(\mu, \nu)$ the set of all measures on $X \times Y$ such that the first marginal is μ and the second marginal is ν . To be precise, if $P^X : X \times Y \rightarrow X$ and $P^Y : X \times Y \rightarrow Y$ are the canonical projections then $P^X_*\pi = \mu$ and $P^Y_*\pi = \nu$. We call any $\pi \in \Pi(\mu, \nu)$ a *transportation plan* between μ and ν (also called a coupling between μ and ν).

The *Kantorovich optimal transport problem* is the following variational problem

$$K(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{X \times Y} c(x, y) d\pi(x, y), \quad (2.1)$$

where $c(x, y)$ is a cost function. The minimiser of this problem is called the *optimal transport plan* $\pi^\dagger$ and such a minimiser exists when c is lower semi-continuous (see, for example [16]). The prototypical example for c (when $X = Y = \mathbb{R}^d$) is $c(x, y) = |x - y|^p := \sum_{i=1}^d |x_i - y_i|^p$, for this choice of c one can define the *Wasserstein distance* by $d_{W^p}(\mu, \nu) = \sqrt[p]{K(\mu, \nu)}$ (see also (2.3) below). When c is a metric then (2.1) is also known as the earth mover's distance.

Now, considering a different formulation, let $T : X \rightarrow Y$ be a Borel measurable function such that $T_*\mu = \nu$. The *Monge optimal transport problem* is to solve

$$M(\mu, \nu) = \inf_{T : T_*\mu = \nu} \int_{X \times Y} c(x, T(x)) d\mu(x). \quad (2.2)$$

We call any T that satisfies $T_*\mu = \nu$ a *transport map* between μ and ν , and the solution to the optimisation problem $T^\dagger$ is called the *optimal transport map*.

It is worthwhile noting that the formulation of the optimal transport problems in Eqs. (2.1) and (2.2) are not, in general, equivalent. However, if the optimal transport plan $\pi^\dagger$ can be written in the form $\pi^\dagger = (\text{Id} \times T^\dagger)_*\mu$ then it follows that $T^\dagger$ is an optimal transport map and the two formulations are equivalent, i.e. $K(\mu, \nu) = M(\mu, \nu)$. A sufficient condition to show that such an optimal transport plan exists is to require that μ is absolutely continuous with respect to the Lebesgue measure on a compact domain $\Omega \subset \mathbb{R}^d$ and, in addition, $c(x, y) = h(x - y)$ where $h : \Omega \rightarrow [0, \infty)$ is strictly convex and superlinear, see [14, Theorem 2.44]. Note that it is easy to find examples where there do not exist transport maps at all. For example, if $\mu = \frac{1}{3}\delta_{x_1} + \frac{1}{3}\delta_{x_2} + \frac{1}{3}\delta_{x_3}$ and $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$.

Let $\Omega \subset \mathbb{R}^n$ and $\mathcal{P}_p(\Omega)$ be the set of Radon measures on Ω with finite p^{th} moment. For $p \in [1, \infty)$, we define the Wasserstein distance between μ and ν in $\mathcal{P}_p(\Omega)$ by

$$d_{W^p}(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\Omega \times \Omega} |x - y|_p^p d\pi(x, y) \right)^{1/p}. \quad (2.3)$$

The Wasserstein space W^p is the metric space $(\mathcal{P}_p(\Omega), d_{W^p})$. For the case $p = \infty$, we can define a distance on $\mathcal{P}_\infty(\Omega)$ by

$$d_{W^\infty}(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \text{ess sup}_\pi \{|x - y| : (x, y) \in \Omega \times \Omega\}.$$

We briefly review the features that make optimal transport particularly suited to signal and image processing. For an extended survey of these ideas see [1]. Optimal transport is able to provide generative models which can represent diverse data distributions and can capture signal variations as a result of spatial perturbations. Furthermore, there is a well formulated theoretical basis (particularly for W^2) with interesting geometrical properties, such as existence of minimisers [15,49], the Riemannian structure of Wasserstein spaces when $p = 2$ [18,50] and characterisation as the weak* convergence when Ω is compact [16]. The Riemannian structure allows the characterisation of geodesics (shortest curves) on the space $\mathcal{P}_2(\Omega)$. In addition, there are many methods to compute optimal transport distances, for convenience, we review a selection in the appendix in the context of computing the TL^p distance.

2.2. The transportation L^p distance

The TL^p distance was first introduced in [42] to define a discrete-to-continuum convergence on point clouds. This tool has been extensively used to study similar statistical problems, e.g. [51–63], and recently has been shown to be a valuable tool in signal analysis [43,64]. In this section, we review the definitions and properties of the TL^p distance and space.

Given an open and bounded domain $\Omega \subset \mathbb{R}^d$ we define the TL^p space as the set of pairs (μ, f) such that $f \in L^p(\mu; \mathbb{R}^m)$ and $\mu \in \mathcal{P}_p(\Omega)$. We do not make any assumption on the dimension m of the range of f . Working in this formal and abstract framework of measure theory allows us to formulate our methods for both discrete and continuous signals, simultaneously. Importantly, similarly to the Wasserstein distance but unlike the L^p distance, this framework allows us to compare signals with different discretisations since μ and ν need not have the same support. We define the TL^p space as

$$TL^p := \{(\mu, f) : \mu \in \mathcal{P}_p(\Omega), f \in L^p(\mu; \mathbb{R}^m)\}.$$

We construct the TL^p metric between pairs $(\mu, f) \in TL^p$ and $(\nu, g) \in TL^p$ as follows:

$$d_{TL^p}((\mu, f), (\nu, g)) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\Omega \times \Omega} |x - y|_p^p + |f(x) - g(y)|_p^p d\pi(x, y) \right)^{1/p}. \quad (2.4)$$

Intuitively, we see that TL^p optimal transport plans (that is plans in $\Pi(\mu, \nu)$ that achieve the minimum in the above variational problem) strike a balance between matching spatially, i.e. minimising $\int_{\Omega \times \Omega} |x - y|_p^p d\pi(x, y)$, and matching signal features, i.e. minimising $\int_{\Omega \times \Omega} |f(x) - g(y)|_p^p d\pi(x, y)$.

Example 2.1. Let us explain here how images can be represented in TL^p . Let $\{x_i\}_{i=1}^n$ be the location of pixels (which usually form a grid over $[0, 1] \times [0, 1]$). We apply TL^p by choosing a base measure μ , this is commonly the uniform measure over $\{x_i\}_{i=1}^n$, i.e. $\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. An image is then represented by the pair (μ, f) where $f : \{x_i\}_{i=1}^n \rightarrow \mathbb{R}^3$ for RGB images and $f(x_i)$ are the RGB values for the pixel at location x_i . Similarly, for greyscale images one would have $f : \{x_i\}_{i=1}^n \rightarrow \mathbb{R}$ where $f(x_i)$ is now the greyscale value for the pixel at location x_i .

Of course, one can make different choices for μ in order to emphasise regions/features of the images. We note that, as we represent the image as a function f , then we do not need to assume that the image has unit mass. To apply W^p we would represent the image as a probability measure μ , and we would therefore have to assume that the image has unit mass (or renormalise). This strong assumption limits the applicability of optimal transport in image processing because it requires ad-hoc renormalisation which may suppress features of the image.

To further understand the TL^p distance, we reformulate it as a W^p distance supported on the graphs of functions. Recall that the graph of a function is defined as:

$$\text{Gra}(f) := \{(x, f(x)) : x \in \Omega\}.$$

Note that the $\text{Gra}(f) \subset \Lambda := \Omega \times \mathbb{R}^m$. We define the following lifted measure on $\text{Gra}(f)$:

$$\tilde{\mu}(A \times B) = (\text{Id} \times f)_* \mu(A \times B) = \mu(\{x : x \in A, f(x) \in B\}),$$

where $A \times B \subset \Lambda$. It is clear that $\tilde{\mu}$ is a well-defined measure on $\text{Gra}(f)$. We can characterise the TL^p distance as a W^p distance in the following way [42]:

$$\begin{aligned} d_{TL^p}^p((\mu, f), (\nu, g)) &= \inf_{\pi \in \Pi(\mu, \nu)} \int_{\Omega \times \Omega} |x - y|_p^p + |f(x) - g(y)|_p^p d\pi(x, y) \\ &= \inf_{\tilde{\pi} \in \Pi(\tilde{\mu}, \tilde{\nu})} \int_{\Lambda \times \Lambda} |x - y|_p^p d\tilde{\pi}(x, y) \\ &= d_{W^p}^p(\tilde{\mu}, \tilde{\nu}). \end{aligned} \quad (2.5)$$

Thus, we can see that the TL^p distance is the W^p distance between the appropriate measures on the graphs of function. This allows us to make the following identification between W^p and TL^p through the mapping.

$$TL^p(\Omega) \rightarrow W^p(\Omega \times \mathbb{R}^m) \quad (2.6)$$

$$(\mu, f) \mapsto \tilde{\mu} = (\text{Id} \times f)_* \mu. \quad (2.7)$$

This connection of the TL^p distance and the W^p distance facilitates the transfer of certain Wasserstein properties to the TL^p setting; for example, metric properties and existence of minimisers.

It is easy to see that, for any $\mu, \nu \in \mathcal{P}_p(\Omega)$ and $f, g \in L^p(\mu)$:

$$\begin{aligned} d_{TL^p}((\mu, f), (\mu, g)) &\leq \|f - g\|_{L^p(\mu)} \\ d_{W^p}(\mu, \nu) &\leq d_{TL^p}((\mu, \mathbb{1}), (\nu, \mathbb{1})). \end{aligned}$$

In fact, one can also prove the converse inequalities (up to a constant) and hence TL^p can be seen to generalise both weak* convergence of measures and L^p convergence of functions (see [42] or Proposition 2.2 below).

The TL^p distance can be seen as a special case of optimal transport by observing that $d_{TL^p}((\mu, f), (\nu, g))$ coincides with the Kantorovich optimal transport problem between two measures μ and ν with cost function $c(x, y; f, g) = |x - y|_p^p + |f(x) - g(y)|_p^p$.

For reference, we also state a Monge-type formulation of the TL^p distance as follows:

$$d_{TL^p}^p((\mu, f), (\nu, g)) = \inf_{T: T_*\mu=\nu} \int_{\Omega} |x - T(x)|_p^p + |f(x) - g(T(x))|_p^p d\mu(x), \quad (2.8)$$

where T is a transportation map. When we write the Monge formulation of optimal transport we are assuming that there is an equivalence between (2.4) and (2.8). This is in general difficult to verify, since the application of Brenier's theorem does not lead to natural conditions. (Assuming that μ does not give mass to small sets and both μ and ν have a sufficient number of bounded moments then one can apply Brenier's theorem to $K(\mu, \nu)$ where $c(x, y) = |x - y|_p^p + |f(x) - g(y)|_p^p$ and K is defined by (2.1), if c is strictly convex; practically this is not reasonable.) However, when μ and ν are discrete uniform measures with supports of equal size the Monge formulation (2.8) coincides with the Kantorovich formulation (2.4) (see the proposition below).

Let us recall the identification (2.6)–(2.7) and the identity $d_{W^p}(\tilde{\mu}, \tilde{\nu}) = d_{TL^p}((\mu, f), (\nu, g))$. Then, there is a corresponding equivalence between

transport maps. That is (assuming all transport maps exist and are unique) let $T^\dagger$ achieve the minimum in (2.8) and $\tilde{T}^\dagger$ achieve the minimum in $M(\tilde{\mu}, \tilde{\nu})$ where M is given by (2.2) with $c(x, y) = |x - y|_p^p$. It follows that $\tilde{T}^\dagger(x) = (T^\dagger(x), g(T^\dagger(x)))$ for μ -almost every $x \in \Omega$.

Proposition 2.2. *Let $\Omega \subset \mathbb{R}^d$ be open and bounded and $p \in (1, \infty)$. Then the following holds*

1. [42, Remark 3.4] $(\text{TL}^p, d_{\text{TL}^p})$ is a metric space;
2. [16, Theorem 5.10] $\mu_n \xrightarrow{*} \mu$ if and only if $(\mu_n, \mathbb{1}) \xrightarrow{\text{TL}^p} (\mu, \mathbb{1})$;
3. [42, Proposition 3.12] $f_n \rightarrow f$ in $L^p(\mu)$ if and only if $(\mu, f_n) \xrightarrow{\text{TL}^p} (\mu, f)$;
4. [43, Proposition 3.4] for any $(\mu, f), (\nu, g) \in \text{TL}^p$ there exists a transport plan $\pi^\dagger \in \Pi(\mu, \nu)$ realising the minimum in $d_{\text{TL}^p}((\mu, f), (\nu, g))$, i.e.

$$d_{\text{TL}^p}^p((\mu, f), (\nu, g)) = \int_{\Omega \times \Omega} |x - y|_p^p + |f(x) - g(y)|_p^p d\pi^\dagger(x, y);$$

5. [43, Proposition 3.5] if $\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\nu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$ then for any $f \in L^p(\mu)$ and $g \in L^p(\nu)$ there exists $T^\dagger : \{x_i\}_{i=1}^n \rightarrow \{y_j\}_{j=1}^n$ such that $T_*^\dagger \mu = \nu$ and

$$d_{\text{TL}^p}^p((\mu, f), (\nu, g)) = \int_{\Omega} |x - T^\dagger(x)|_p^p + |f(x) - g(T^\dagger(x))|_p^p d\mu(x),$$

i.e. the Monge and Kantorovich formulations of TL^p (given by (2.4) and (2.8) respectively) are equivalent for point masses.

Note that not all properties of W^p carry through to TL^p . For example $(\text{TL}^p, d_{\text{TL}^p})$ is not complete. Indeed, following [42], let $\Omega = (0, 1)$ and note that $f_{n+1}(x) = \text{sign} \sin(2^n \pi x)$, $\mu_n = \mathcal{L}|_{(0,1)}$ (the Lebesgue measure on $(0, 1)$) is a Cauchy sequence in $(\text{TL}^p, d_{\text{TL}^p})$. However, $\{f_n\}$ does not converge in L^p and therefore $\{(\mu_n, f_n)\}$ cannot converge in TL^p , by part 3 of the above proposition. The completion of TL^p can be identified with the set of Young measures, and therefore the space $\mathcal{P}(\Omega \times \mathbb{R}^m)$, see [42, Remark 3.6]. We note also that there do not exist geodesics in TL^p , however we develop an approach to interpolate in TL^p (see Section 2.5).

By part 3 in the above proposition TL^p inherits some sensitivity to high frequency perturbations from the L^p norm. In contrast, as W^p metricises the weak* convergence (in compact Euclidean spaces) then W^p is insensitive to high frequency perturbations, see [43, Section 2.2].

We also can deduce that the TL^p distance inherits translation sensitivity from W^p . In particular, we can see that L^p distances are insensitive to translations if the supports of the images are disjoint. On the other hand, the W^p distance scales linearly with the size of the translation no matter how large the translation. The TL^p distance, although not scaling linearly with translation, is monotonically non-decreasing as a function of translation.

The TL^p appears an excellent tool to exploit in pattern recognition problems such as images or times series, however it is as computational demanding as W^p and despite recent advances in computation of optimal transport, e.g. [30], it is still challenging to apply it to large scale problems. In the next section, we review the linear Wasserstein (LW^p) framework, which was introduced to allow application of optimal transport methods to classification problems [38]. In later sections, we apply the ideas of linear optimal transport to TL^p .

2.3. A linear W^p framework

The LW^p framework was proposed in [38], as a way to apply optimal transport techniques (in particular W^p) to large scale classification problems for image analysis. Given a set of N images, one would need to compute all pairwise W^p distances in order to use methods such as k -nearest neighbour classifiers. The LW^p framework was developed so that the Wasserstein distance needs to be computed only N times. In particular, it is the optimal Wasserstein transport maps between signals

that are computed. From here, the images are embedded in Euclidean space therefore allowing linear statistical techniques to be applied [65]. This technique was successfully applied in [2] to detect morphological difference in cancer cells. The technique has been further refined and extended to super-resolution images [3,39]. In this section, we briefly review the ideas of LW^p .

The idea behind the LW^p framework is to find the optimal Wasserstein transport maps with respect to one (reference) measure. For simplicity we assume that the Monge problem is equivalent to the Kantorovich problem and, in particular, there exists optimal transport maps. Via an embedding of the transport map into Euclidean space the W^p distance between any two pairs is estimated. More precisely, the LW^p framework provides a linear embedding for $\mathcal{P}_p(\Omega)$ with respect to a fixed measure $\sigma \in \mathcal{P}_p(\Omega)$ [1]. This means the Euclidean distance of the embedded measure and the fixed measure σ is equal to the W^p distance of the measure and the fixed measure. The Euclidean distance between any two measures is then an approximation to the Wasserstein distance between these measures. These linear embeddings then facilitate the application of standard statistical techniques such as PCA, LDA and K-means. The LW^p framework is also invertible and so synthetic, but physically possible signals, can be realised [40].

Let $\mu_1, \mu_2 \in \mathcal{P}(\Omega)$ and $\sigma \in \mathcal{P}(\Omega)$ is our reference measure. Throughout this section, we assume that optimal transport maps $T^{\mu_i} : \Omega \rightarrow \Omega$ exist between σ and μ_i , i.e.

$$d_{W^p}(\sigma, \mu_i) = \sqrt[p]{\int_{\Omega} |x - T^{\mu_i}(x)|_p^p d\sigma(x)},$$

and $T_*^{\mu_i} \sigma = \mu_i$. If optimal transport maps do not exist then one can still define the LW^p distance but there is not a natural way to embed this distance into Euclidean space. We refer to [38, Section 2.3] on how to define the LW^p distance using generalised geodesics.

In the setting considered here the *Linear Wasserstein Distance*, LW^p , is defined by [38]:

$$d_{LW^p, \sigma}(\mu_1, \mu_2) := \sqrt[p]{\int_{\Omega} |T^{\mu_1}(x) - T^{\mu_2}(x)|_p^p d\sigma(x)}. \quad (2.9)$$

We observe that $d_{LW^p, \sigma}$ is a metric and

$$d_{LW^p, \sigma}(\mu_1, \mu_2) = \|T^{\mu_1} - T^{\mu_2}\|_{L^p(\sigma)}.$$

Let us assume that σ has a density ρ with respect to the Lebesgue measure and define

$$P_c(\mu) = (T^\mu - \text{Id})\rho^{\frac{1}{p}}. \quad (2.10)$$

Then

$$d_{LW^p, \sigma}(\mu_1, \mu_2) = \|P_c(\mu_1) - P_c(\mu_2)\|_{L^p(\Omega)}. \quad (2.11)$$

The map P_c is our linear embedding from the Wasserstein space to Euclidean space. We make the following claims on the embedding.

Proposition 2.3. *Assume $\Omega \subset \mathbb{R}^d$ is bounded and $\sigma \in \mathcal{P}(\Omega)$ has a density ρ with respect to the Lebesgue measure. Define $P = P_c$ where P_c is given by (2.10). Then, the following holds:*

1. $P(\mu) \in L^p(\Omega)$ for any $\mu \in \mathcal{P}(\Omega)$,
2. $P(\sigma) = 0$,
3. $d_{LW^p, \sigma}(\mu_1, \mu_2) = \|P(\mu_1) - P(\mu_2)\|_{L^p(\Omega)}$ for any $\mu_1, \mu_2 \in \mathcal{P}(\Omega)$,
4. $d_{LW^p, \sigma}(\sigma, \mu) = d_{W^p}(\sigma, \mu)$ for any $\mu \in \mathcal{P}(\Omega)$,
5. if in addition $\rho = 1$ on Ω there exists a constant C (independent of μ, ν and σ and p and depending only on Ω) such that

$$d_{W^p}(\mu, \nu) \leq d_{LW^p, \sigma}(\mu, \nu) \leq C d_{W^p}(\mu, \nu)^{\frac{2}{15 \min\{1, \frac{2}{p}\}}}.$$

Proof. Since σ has a density with respect to the Lebesgue measure then transport maps T^{μ_i}, T^μ exist. Since the W^p distance is finite (as Ω

is bounded), it follows that $T^{\mu_i} - \text{Id} \in L^p(\sigma)$ which proves (1). Claim (2) follows directly from $T^\sigma = \text{Id}$. Claim (3) was shown already in (2.11). Claim (4) follows

$$d_{LW^p, \sigma}(\sigma, \mu) = \|(T^\mu - \text{Id})\rho_p^{\frac{1}{p}}\|_{L^p(\Omega)} = \|T^\mu - \text{Id}\|_{L^p(\sigma)} = d_{W^p}(\sigma, \mu)$$

where we use $P(\sigma) = 0$. Finally, claim (5) follows from [66], in particular the lower bound is immediate and the upper bound is an application of [66, Theorem 3.1] and Jensen's inequality for $p \in [1, 2]$:

$$d_{LW^p, \sigma}(\mu, \nu) = \|T^\mu - T^\nu\|_{L^p(\sigma)} \leq \|T^\mu - T^\nu\|_{L^2(\sigma)} \leq C_1 d_{W^p}(\mu, \nu)^{\frac{2}{15}};$$

and [66, Theorem 3.1] with a straightforward bound on L^p norms for $p \geq 2$:

$$\begin{aligned} d_{LW^p, \sigma}(\mu, \nu) &\leq \text{diam}(\Omega)^{\frac{p-2}{p}} \|T^\mu - T^\nu\|_{L^2(\sigma)}^{\frac{2}{p}} \\ &\leq C_2 \max\{1, \text{diam}(\Omega)\} d_{W^p}(\mu, \nu)^{\frac{4}{15p}}. \quad \square \end{aligned}$$

We make a similar definition for discrete measures. If $\sigma = \sum_{j=1}^n \rho_j \delta_{x_j}$ for some $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ then we define

$$[P_d(\mu_i)]_j = (T^{\mu_i}(x_j) - x_j)\rho_j^{\frac{1}{p}}. \quad (2.12)$$

Analogously to the Lebesgue density case we have

$$d_{LW^p, \sigma}(\mu_1, \mu_2) = |P_d(\mu_1) - P_d(\mu_2)|_p$$

where we recall that $|\cdot|_p$ is the Euclidean p -norm: $|x|_p := \sqrt[p]{\sum_{j=1}^n |x_j|^p}$. For discrete σ the map P_d is our linear embedding from the Wasserstein space to Euclidean space. We have the analogue of Proposition 2.3 in the discrete case; however, since Proposition 2.3.5 relied upon the bounds in [66] which are specific to reference measures with continuous densities, we no longer obtain a bound between LW^p and W^p . It is beyond the scope of this paper to extend the results of [66] to more general reference measures.

Proposition 2.4. Let $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ and assume $\sigma = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. Define $P = P_d$ where P_d is given by (2.12) with $\rho_j = \frac{1}{n}$. Then, the following holds:

1. $P(\mu) \in \mathcal{C}^p$ for any $\mu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$,
2. $P(\sigma) = 0$,
3. $d_{LW^p, \sigma}(\mu_1, \mu_2) = |P(\mu_1) - P(\mu_2)|_p$ for any $\mu_1 = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$, $\mu_2 = \frac{1}{n} \sum_{j=1}^n \delta_{z_j}$,
4. $d_{LW^p, \sigma}(\sigma, \mu) = d_{W^p}(\sigma, \mu)$ for any $\mu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$.

Proof. The particular forms of all the measures $\sigma, \mu, \mu_1, \mu_2$ is enough to guarantee that transport maps $T^\mu, T^{\mu_1}, T^{\mu_2}$ all exist. The proof is then analogous to the proof of Proposition 2.3. $\square$

In both the Lebesgue density and uniform discrete case P preserves the W^p distance between the reference measure and any given μ (where for discrete measures μ is also uniform discrete). Between μ_1, μ_2 one approximates $d_{W^p}(\mu_1, \mu_2) \approx d_{LW^p, \sigma}(\mu_1, \mu_2)$. The next section proposes our extension of the LW^p framework to the TL^p distance. We conclude this section with a remark on the choice of reference.

Remark 2.5. Since the Wasserstein space is curved then the choice of reference measure affects the projections (the linearisation is the projection onto the tangent space, and since the space is curved the tangent spaces do not coincide). Therefore how well the linear Wasserstein distance approximates the Wasserstein distance depends on the choice of reference. For a data set $\{\mu_i\}_{i=1}^n$ the theoretical best choice is the Wasserstein barycentre. However, the decrease in the quality of the approximation depends on the curvature in the region of the data, which in the worst case is Hölder continuous with exponent $\frac{2}{15}$ [66] for $p = 2$, i.e.

$$d_{W^2}(\mu_1, \mu_2) \leq d_{LW^2}(\mu_1, \mu_2) \leq C d_{W^2}(\mu_1, \mu_2)^{\frac{2}{15}}$$

and in general not better than Hölder continuous with exponent $\frac{1}{p}$ [67]. However for certain classes of measures better bounds exist, for example if the data consists of small perturbations of shifts and scalings [68]. In our numerical experiments, although our data does not fit the theory in [68] we find that the results are not sensitive to the choice of reference. We observe similar results when linearising the TL^p distance.

2.4. A linear TL^p framework

In this section, we propose a linear TL^p framework. Recall that the TL^p distance can be defined as an optimal transport distance between measures supported on the graph of a function. Let $(\sigma, h) \in TL^p$ be the TL^p reference signal and $\tilde{\sigma} = (\text{Id} \times h)_* \sigma \in \mathcal{P}_p(\Lambda)$ the measure in $\Lambda = \Omega \times \mathbb{R}^m$ with support on the graph of h . Let $(\mu_i, f_i) \in TL^p$, $i = 1, 2$, and define $\tilde{\mu}_i = (\text{Id} \times f_i)_* \mu_i$. As in the previous section we will assume that optimal transport maps $\tilde{T}^{\tilde{\mu}_i} : \Lambda \rightarrow \Lambda$ exist between $\tilde{\sigma}$ and $\tilde{\mu}_i$, i.e.

$$d_{W^p}(\tilde{\sigma}, \tilde{\mu}_i) = \sqrt[p]{\int_{\Lambda} |x - \tilde{T}^{\tilde{\mu}_i}(x)|_p^p d\tilde{\sigma}(x)}.$$

Recall that we can write $\tilde{T}^{\tilde{\mu}_i}$ in the form $\tilde{T}^{\tilde{\mu}_i}(x) = (T^{\mu_i}(x), f_i(T^{\mu_i}(x)))$ where $x = (x, y) \in \mathbb{R}^d \times \mathbb{R}^m$ and T^{μ_i} is the optimal plan for the Monge problem (2.2) between μ_i and σ with cost $c(x, y) = |x - y|_p^p + |f(x) - g(y)|_p^p$. The *Linear Transportation L^p Distance* (LTL^p) is defined as

$$d_{LTL^p, (\sigma, h)}((\mu_1, f_1), (\mu_2, f_2)) := \sqrt[p]{\int_{\Lambda} |\tilde{T}^{\tilde{\mu}_1}(x) - \tilde{T}^{\tilde{\mu}_2}(x)|_p^p d\tilde{\sigma}(x)}. \quad (2.13)$$

Simple manipulations of the LTL^p distance imply

$$\begin{aligned} d_{LTL^p, (\sigma, h)}((\mu_1, f_1), (\mu_2, f_2)) &= \|\tilde{T}^{\tilde{\mu}_1} - \tilde{T}^{\tilde{\mu}_2}\|_{L^p(\tilde{\sigma})} \\ &= \sqrt[p]{\int_{\Omega} |T^{\mu_1}(x) - T^{\mu_2}(x)|_p^p + |f_1(T^{\mu_1}(x)) - f_2(T^{\mu_2}(x))|_p^p d\sigma(x)}. \end{aligned}$$

Comparing (2.9) and (2.13) we can see that LTL^p generalises LW^p in the same way that TL^p generalises W^p . Indeed, whereas the LW^p is the L^p norm of transport maps with respect to the measure σ , the LTL^p distance is the L^p norm of transport maps with respect to σ raised to the graph of h .

Following the construction of the embedding in the previous section we go directly to the discrete case (since $\tilde{\sigma}$ has support on the graph it cannot have a density with respect to the Lebesgue measure). We assume that $\sigma = \sum_{i=1}^n \rho_j \delta_{x_j}$ for some $\{x_j\}_{j=1}^n \subset \mathbb{R}^d$ and we define

$$[P_d((\mu_i, f_i))]_j = (T^{\mu_i}(x_j) - x_j)\rho_j^{\frac{1}{p}} \quad (2.14)$$

$$[Q_d((\mu_i, f_i))]_j = (f_i(T^{\mu_i}(x_j)) - h(x_j))\rho_j^{\frac{1}{p}} \quad (2.15)$$

$$\tilde{P}_d((\mu_i, f_i)) = (P_d((\mu_i, f_i)), Q_d((\mu_i, f_i))). \quad (2.16)$$

Given this definition we can write

$$d_{LTL^p, (\sigma, h)}((\mu_1, f_1), (\mu_2, f_2)) = |\tilde{P}_d((\mu_1, f_1)) - \tilde{P}_d((\mu_2, f_2))|_p.$$

The map $\tilde{P}_d$ embeds our signals into Euclidean space. We have the following properties of the embedding (analogous to Proposition 2.4). Note that Proposition 2.3 assumed that the reference measure has a density. To pass theoretical results from W^p in the continuous density case to TL^p via the identity (2.5) would require the assumption that $\tilde{\sigma}$ has a continuous density. As $\tilde{\sigma}$ is supported on the graph of a function this is clearly not true and so the results do not immediately follow. However, we believe this is just a technical difficulty and we would expect the analogous result to Proposition 2.3 to hold for TL^p (in particular, the bound on the approximation in part (5)).

Proposition 2.6. Let $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ and assume $\sigma = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. Define $\tilde{P} = \tilde{P}_d$ where $\tilde{P}_d$ is given by (2.14)–(2.16) with $\rho_j = \frac{1}{n}$. Then, the following holds:

1. $\tilde{P}((\mu, f)) \in \ell^p$ for any $(\mu, f) \in \text{TL}^p$ with $\mu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$,
2. $P((\sigma, h)) = 0$,
3. $d_{\text{TL}^p}((\mu_1, f_1), (\mu_2, f_2)) = |\tilde{P}((\mu_1, f_1)) - \tilde{P}((\mu_2, f_2))|_p$ for any $(\mu_i, f_i) \in \text{TL}^p$ with $\mu_1 = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$, $\mu_2 = \frac{1}{n} \sum_{j=1}^n \delta_{z_j}$,
4. $d_{\text{TL}^p}((\sigma, h), (\mu, f)) = d_{\text{TL}^p}((\sigma, h), (\mu, f))$ for any $(\mu, f) \in \text{TL}^p$ with $\mu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$.

Proof. By Proposition 2.2(5) the transport maps $T^\mu, T^{\mu_1}, T^{\mu_2}$ exist. The rest of the proof follows as in the proof of Proposition 2.3. $\square$

As for LW^p we have that LTL^p is exactly TL^p when comparing with the reference measure, i.e. $d_{\text{LTL}^p, (\sigma, h)}((\sigma, h), (\mu, f)) = d_{\text{TL}^p}((\sigma, h), (\mu, f))$. When we are comparing two measures, neither of which are the base measure, then we make the approximation

$$d_{\text{LTL}^p, (\sigma, h)}((\mu_1, f_1), (\mu_2, f_2)) \approx d_{\text{TL}^p}((\mu_1, f_1), (\mu_2, f_2)).$$

When $(\mu_i, f_i) \neq (\sigma, h)$ then the approximation is only formal.

2.5. Inverting the linear embedding

So far our motivation has been to define an embedding so that the TL^p distance between (μ_1, f_1) and (μ_2, f_2) is approximated by a Euclidean distance between $\tilde{P}_d((\mu_1, f_1))$ and $\tilde{P}_d((\mu_2, f_2))$. Our interest in creating this embedding was partly numerical, since computing pairwise distance on the data set $\{\tilde{P}_d((\mu_i, f_i))\}_{i=1}^n$ is quicker than computing the TL^p distance on the data set $\{(\mu_i, f_i)\}_{i=1}^n$, and partly to allow the application of Euclidean data analysis techniques. For the latter it is beneficial to be able to invert the mapping. For example, one can consider fitting models for data generation in TL^p (following the same method for the Wasserstein distance [40]) or interpolating in the Euclidean space and pulling the interpolation back to the TL^p space. The aim of this section is to define the inverse $\tilde{P}_d^{-1}$ which maps from a Euclidean space back into TL^p .

The space $(P_p(\Omega), d_{\text{WP}})$ is a geodesic space, with easily characterisable geodesics. Letting $\pi^\dagger \in \Pi(\sigma, \mu)$ be the optimal transport plan that minimises the transport problem given by (2.1), we define $I_t : \Omega \times \Omega \rightarrow \Omega$, where $t \in [0, 1]$, to be a linear interpolation, as follows:

$$I_t(x, y) = (1 - t)x + ty.$$

Then the geodesic in W^p is given by $\mu(t) = [I_t]_* \pi^\dagger$. When there exists transport maps, i.e. $\pi^\dagger = (\text{Id} \times T^\mu)_* \mu$ then the geodesic can be written $\mu_t = [T_t^\mu]_* \mu$ where $T_t^\mu(x) = I_t(x, T^\mu(x)) = (1 - t)x + tT^\mu(x)$.

What this means is that if we consider any map $T : \Omega \rightarrow \Omega$ that is in the convex hull of $\{T^{\mu_i}\}_{i=1}^n$ then defining $\mu = T_* \sigma$ implies that $T = T^\mu$. Hence, the inverse linear Wasserstein mapping can be defined. In the continuum case we may define (for $v \in L^p(\Omega)$ in the convex hull of $\{P_c(\mu_i)\}_{i=1}^n$)

$$P_c^{-1}(v) = T_* \sigma \quad \text{where } T(x) = v(x)\rho(x)^{-\frac{1}{p}} + x$$

under the assumptions of Proposition 2.3. And in the discrete case we may define (for $v \in \ell^p$ in the convex hull of $\{P_d(\mu_i)\}_{i=1}^n$)

$$P_d^{-1}(v) = T_* \sigma \quad \text{where } T(x_j) = n^{\frac{1}{p}} v_j + x_j$$

under the assumptions of Proposition 2.4.

Since,

$$P(\mu_t) = (T_t^\mu - \text{Id})\rho^{\frac{1}{p}} = ((1 - t)\text{Id} + tT^\mu - \text{Id})\rho^{\frac{1}{p}} = t(T^\mu - \text{Id})\rho^{\frac{1}{p}} = tP(\mu)$$

where $P = P_c$ or $P = P_d$ then we see that the projection of the geodesic onto the Euclidean space is the geodesic between the projections. In particular, the geodesic between $P(\sigma) = 0$ and $P(\mu)$ in Euclidean space is simply $tP(\mu)$. Notably, this allows one to translate principal eigenvectors in PCA space (of the linear embedding) into modes of variation in W^p , see [38] for more details.

Note that to generate the new image we only required the reference measure σ and a new point v in the linear space. However, in order to generate the new point v it will often be sensible to use the statistics of $\{P(\mu_i)\}_{i=1}^N$, for example see [40].

This argument does not directly apply to the TL^p space since, by the following remark, the TL^p space does not permit geodesics.

Remark 2.7. Let $(\mu, f), (v, g)$ be in TL^p and $\tilde{\mu} = (\text{Id} \times f)_* \mu$, $\tilde{v} = (\text{Id} \times g)_* v$. If $\{(\mu_t, f_t)\}_{t \in [0, 1]}$ is the TL^p shortest curve between (μ, f) , (v, g) then (via (2.5)) we can see that $\tilde{\mu}_t = (\text{Id} \times f_t)_* \mu_t$ is the shortest curve in W^p . Since the geodesic in W^p exists and is unique then the converse statement is that if $\tilde{\mu}_t$ is the W^p shortest curve between $\tilde{\mu}$ and $\tilde{v}$ and $\tilde{\mu}_t$ can be written in the form $\tilde{\mu}_t = (\text{Id} \times f_t)_* \mu_t$ then $(\mu_t, f_t)_*$ is the shortest curve in TL^p . It is this second condition, that $\tilde{\mu}_t$ can be written in the form $\tilde{\mu}_t = (\text{Id} \times f_t)_* \mu_t$, that can fail. For example, consider the measure $\mu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and the functions $f(0) = 0, f(1) = 10, g(0) = 10, g(1) = 0$. Then the transport between (μ, f) and (μ, g) is from $(0, 0)$ to $(1, 0)$ and from $(1, 10)$ to $(0, 10)$. The "half way" point in W^p would be the measure $\tilde{\mu}_{\frac{1}{2}} = \frac{1}{2}\delta_{(\frac{1}{2}, 0)} + \frac{1}{2}\delta_{(\frac{1}{2}, 10)}$. If it exists the halfway point in TL^p would be the measure $\mu_{\frac{1}{2}}$ and a function $f_{\frac{1}{2}}$ such that $(\text{Id} \times f_{\frac{1}{2}})_* \mu_{\frac{1}{2}} = \frac{1}{2}\delta_{(\frac{1}{2}, 0)} + \frac{1}{2}\delta_{(\frac{1}{2}, 10)}$. But since this would require the function $f_{\frac{1}{2}}$ to take both the value 0 and 10 at $x = \frac{1}{2}$ then it is clear no such function can exist.

However, this does not prevent us from interpolating and visualising modes of variation. Indeed, let $\tilde{T}^\mu(x) = (T^\mu(x), f(T^\mu(x)))$ be the optimal TL^p map pushing (σ, h) to (μ, f) , then the map

$$\tilde{T}_t^\mu(x) = ((1 - t)x + tT^\mu(x), (1 - t)h(x) + t(f(T^\mu(x))))$$

interpolates between the signals (σ, h) and (μ, f) . In fact, this is the geodesic in $\text{W}^p(\Omega \times \mathbb{R}^m)$; that is $\tilde{\mu}_t = [\tilde{T}_t^\mu]_* \tilde{\sigma}$ is the geodesic in $\text{W}^p(\Omega \times \mathbb{R}^m)$ between $\tilde{\sigma}$ and $\tilde{\mu}$. Although we can invert $\tilde{P} = \tilde{P}_d$ (defined in (2.14)) in the Wasserstein space, i.e. for all $p \in \mathbb{R}^m$ there exists $\tilde{v} \in P(\Omega \times \mathbb{R}^m)$ such that $P(\tilde{v}) = p$ (note that this is P and not $\tilde{P}$ since we are inverting with respect to $\text{W}^p(\Omega \times \mathbb{R}^m)$) we cannot guarantee that $\tilde{v}$ can be written in the form $\tilde{v} = (\text{Id} \times g)_* v$ for some $g \in L^p(v)$. Hence, we cannot in general invert the linear embeddings from TL^p back into TL^p . Instead we use an approximate inversion. We define $\tilde{P}^{-1}(p) = (v, \tilde{g})$ where $\tilde{v}$ satisfies $P(\tilde{v}) = p$, $\tilde{g}(x) = \mathbb{E}_{\tilde{v}_x}$ and (where we use disintegration of measures) $\tilde{v} = \tilde{v}_x \otimes v$ with the latter meaning

$$\tilde{v}(A \times B) = \int_A \tilde{v}_x(B) d\nu(x) \quad \text{for all measurable } A \subset \Omega, B \subset \mathbb{R}^m.$$

In other words, we define the "inverse" map from the linear embedding of TL^p back into TL^p as the inverse map in $\text{W}^p(\Omega \times \mathbb{R}^m)$ and projected onto $\text{TL}^p(\Omega)$:

- 1: $p \in \{\text{linear TL}^p \text{ space}\} \mapsto \tilde{v} \in P(\Omega \times \mathbb{R}^m)$ using inverse of LW^p

in the space $\Omega \times \mathbb{R}^m$

- 2: $\tilde{v} \mapsto (v, g) \in \text{TL}^p$ by projecting $\text{W}^p(\Omega \times \mathbb{R}^m)$ onto $\text{TL}^p(\Omega)$.

The projection onto TL^p is done by taking the mean across each fibre in $x \in \Omega$. We note that the second step does not have to be performed in the linear Wasserstein setting. With this definition we are also able to visualise any point in the linear embedding in TL^p space.

Example 2.8. Let us consider how to generate a new TL^p colour image from the linear space. We recall that colour images can be represented by (μ, f) where $\{x_i\}_{i=1}^n$ are the locations of pixels (which are uniform across $[0, 1] \times [0, 1]$), $\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $f : \{x_i\}_{i=1}^n \rightarrow \mathbb{R}^3$ represents the RGB values for each pixel. We take a reference image (σ, h) of the same form (in particular $\sigma = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$), and note that the embedding (mapping $\tilde{\sigma} = (\text{Id} \times h)_* \sigma$ to $\tilde{\mu} = (\text{Id} \times f)_* \mu$) is given by $\alpha^{\tilde{\mu}} = (\alpha_1^{\tilde{\mu}}, \dots, \alpha_n^{\tilde{\mu}}) \in \mathbb{R}^{5n}$ where

$$\alpha_i^{\tilde{\mu}} = \frac{1}{n^p} (\tilde{T}^\mu(x_i) - (x_i, h(x_i))) \in \mathbb{R}^5.$$

To generate new images we need to invert this mapping. Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{R}^{5n}$. We define $\tilde{T}_i = n^{\frac{1}{p}} \alpha_i + (x_i, h(x_i)) \in \mathbb{R}^5$. Then $y_i = (\tilde{T}_i)_{1:2} \in \mathbb{R}^2$ are the location of the pixels and $c_i = (\tilde{T}_i)_{3:5}$ are the RGB values in the new image. In the TL^p space the new image is represented by (v, g) where $v = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$ and $g(y_i) = c_i$. This is only well defined if y_i are all unique. If not, then we use *barycentric projection* (as we also do in Section 2.5), for example if $y_i = y$ for all $i \in I$ then we define $g(y) = \frac{1}{|I|} \sum_{i \in I} c_i$ to be the empirical average.

3. Results

In this section we apply the LTL^p framework to three real world examples to auslan (Australian sign language), breast cancer histopathology and financial time series. We also include two synthetic examples and a further application to cell morphometry in the appendix. Throughout we will choose $p = 2$ and we use the Euclidean mean signal for the reference. Empirically, we find that the choice of reference does not have a significant effect on our results. Therefore, although the W^p and TL^p barycentre will give the smallest average error in the linear approximations, we use the easy to compute Euclidean average (for both LW^p and LTL^p).

We use the popular entropy regularised approach to computing the TL^p -optimal maps first proposed by [30]. In particular, adding entropic regularisation allows one to rewrite the optimal transport problem (in our case TL^p distance) as a Kullback–Liebler divergence. We use the Sinkhorn algorithm to solve entropy regularised approximation using the algorithm based on [32]. The entropy regularised solution causes blurring in the optimal plan which means that the support of the optimal plan is not supported on the graph of a function (i.e. in the notation of Proposition 2.2 there does not exist $T^\dagger$ such that $\pi^\dagger = (\text{Id} \times T^\dagger)_\# \mu$). To overcome this, following [19] we use barycentric projection to approximate the optimal $\pi^\dagger$, as found by solving the entropic regularised formulation, by a map which is supported on the graph of a function. For completeness more details on the numerical scheme are given in Appendix A.

3.1. Application to Auslan data

We apply the transportation methodology presented in this manuscript to the Australian Sign language (Auslan) dataset [69]. A native Auslan signer was recorded, using fifth dimension technology gloves, making 95 different signs repeated over a period of 9 weeks. The sign was repeated 3 times at each recording, thus each word was measured 27 times. This means there are a total of 2565 signs in the dataset. Each measurement is considered as a multivariate time-series. The measurements taken for each hand are the x, y, z positions, along with roll, pitch and yaw. In addition, the bend of each of the 5 fingers is recorded. Thus at each frame 22 measurements are observed. We consider the Auslan data as functions $f_i : \{t_1, t_1, \dots, t_M\} \rightarrow \mathbb{R}^{22}$, for $i = 1, \dots, 2565$ and where $\{t_1, \dots, t_M\}$ are the times. We truncate the number of time frames to $M = 44$ because little variation was observed past this point. The goal of this task is to classify signs given $f_i : \{t_1, \dots, t_{44}\} \rightarrow \mathbb{R}^{22}$, $i = 1, \dots, 2565$ as input to 95 possible words (labels) as output.

We apply the LTL^p and LW^p frameworks to this dataset. Since TL^p can handle multi-channel signals, no additional pre-processing was needed. However, to apply W^p additional pre-processing was required. Firstly, all signals were made positive and then the mean was taken so that there was only a signal channel with positive values. We then normalised so the signal integrated to unity. A linear embedding was obtained as described in previous sections. Once this linear embedding is obtained, we use the 1 nearest neighbour¹ (1NN) algorithm to predict

Table 1

CPU times in seconds to compute each transportation method on each dataset. Computation was halted after 4 CPU days (= 345600 s).

Application	LW^p	LTL^p	TL^p
Auslan	12.1	13.0	91 200
Breast Cancer Histopathology	25 407.0	2919.8	>345600
Financial Time Series	39.5	192.3	–

the signs from the linear embedding of the signals. As an assessment of performance we use the macro-F1 score.² We assess performance with a 5-fold cross-validation framework and repeat 100 times to produce a distribution of scores. In addition, we compare to the standard TL^p methodology since for this particular data set, even though costly it is possible to perform the full computation. We also recorded timings for each of the methods.

Table 1 shows that the linear transportation methods are considerably faster than the full transportation methods. Indeed the LTL^p distance was on the order of magnitude of ten's of seconds, whilst the full TL^p took several hours. Fig. 1 demonstrates that our proposed LTL^p method significantly outperforms the LW^p approach (T-test, $p < 10^{-4}$) on the Auslan data. There is a loss in classification ability of LTL^p versus the TL^p method, which is unsurprising as the linear transportation method is approximate. However, on the Auslan dataset we observe this difference to be small and this minor improvement comes at computational cost orders of magnitude greater.

3.2. Application to breast cancer histopathology

In this section, we demonstrate the applicability of LTL^p to images from breast cancer histopathology. Invasive Ductal Carcinoma is an aggressive and common form of breast cancer and deep learning based approaches have been used to construct classifiers to analyse such data [71,72]. We analyse these datasets using transportation based approaches. We randomly sample 100 images each from two patients 1 healthy and 1 cancerous, totalling 200 images. Each image is on a 50×50 pixel grid. To apply LW^p to these images we first convert the images to a single intensity channel and renormalise so that the intensities integrate to unity. We apply the LTL^p approach without any ad-hoc preprocessing, since it can be directly applied to un-normalised multi-channelled images. We compute transport maps in each case using entropy regularised approaches and then linearly embed these images as described in earlier sections. We visualise the linear embeddings using PCA. In addition, we again employ the 1NN classifier using the same framework as in the Auslan application. We report distributions of macro-F1 scores for both LW^p and LTL^p .

From Table 1 we see that the computational cost of the LTL^p and LW^p distances differs significantly more than might be expected. This is due to the LTL^p distance converging more quickly, requiring on an average of 10 (up to a maximum of 100) iterations compared to an average of 100 (up to a maximum of 200) iterations for LW^p in our Sinkhorn implementation. The TL^p distance was not included as a comparison as it was too expensive to compute.

Fig. 2 demonstrates clear differences between the LTL^p embedding and LW^p embedding. Comparing Figs. 2(a) and (b) we see that the

¹ Our 1NN classifier using 5-fold cross validation uses 80% of the data with labels and the remaining 20% without labels. We predict the labels on the 20% by finding the nearest signal (in the appropriate distance) in the 80% that are labelled and using that label. I.e. if f_i is unlabelled and f_j is the closest labelled signal then we use f_j 's label as our estimate for the label of f_i .

² The macro-F1 score [70] is the harmonic mean of the precision and recall. The precision is defined as the fraction of true positives to all positives, i.e. if we consider the first word we would look at the fraction of signals we correctly identified as belonging to this class to all the signals we put in this class. The recall is defined as the fraction of signals in each class that were correctly identified, i.e. we find the accuracy of the method looking only at the signals where the true label is the first word, then repeat for the second, third, fourth, etc words. An F1-score of 1 would imply perfect precision and recall, whilst a low F1 score would imply either or both of the precision and recall were performing poorly.

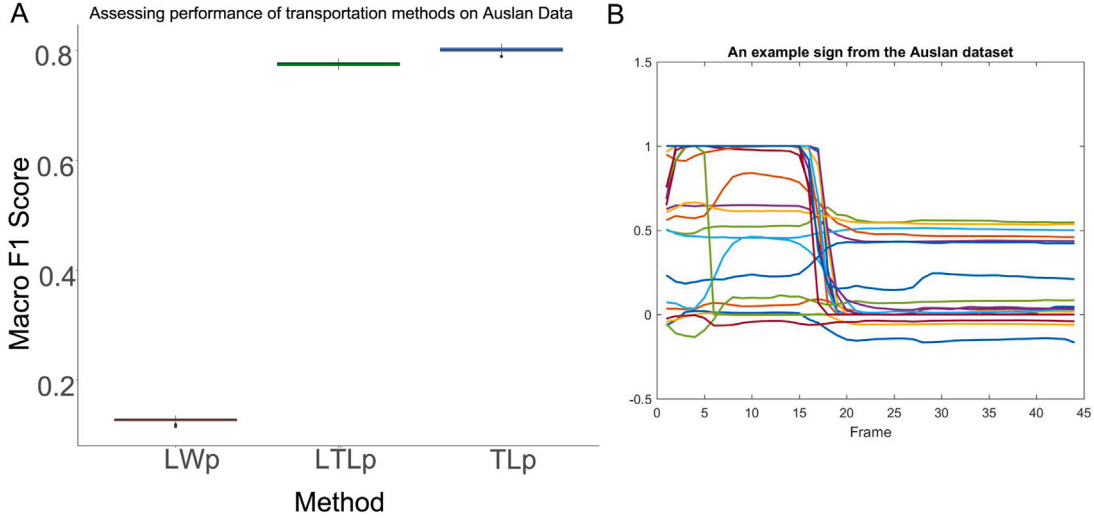


Fig. 1. (A) Distributions of macro-F1 scores for different transportation methods. Evidently LTL^p significantly outperforms LW^p . (B) An example a multi-variate signal $f_i : \{t_1, \dots, t_{44}\} \rightarrow \mathbb{R}^{22}$ from the Auslan dataset. Each curve represents one of the 22 sensors. We see that the informative part of the signal is before the 20th time frame.

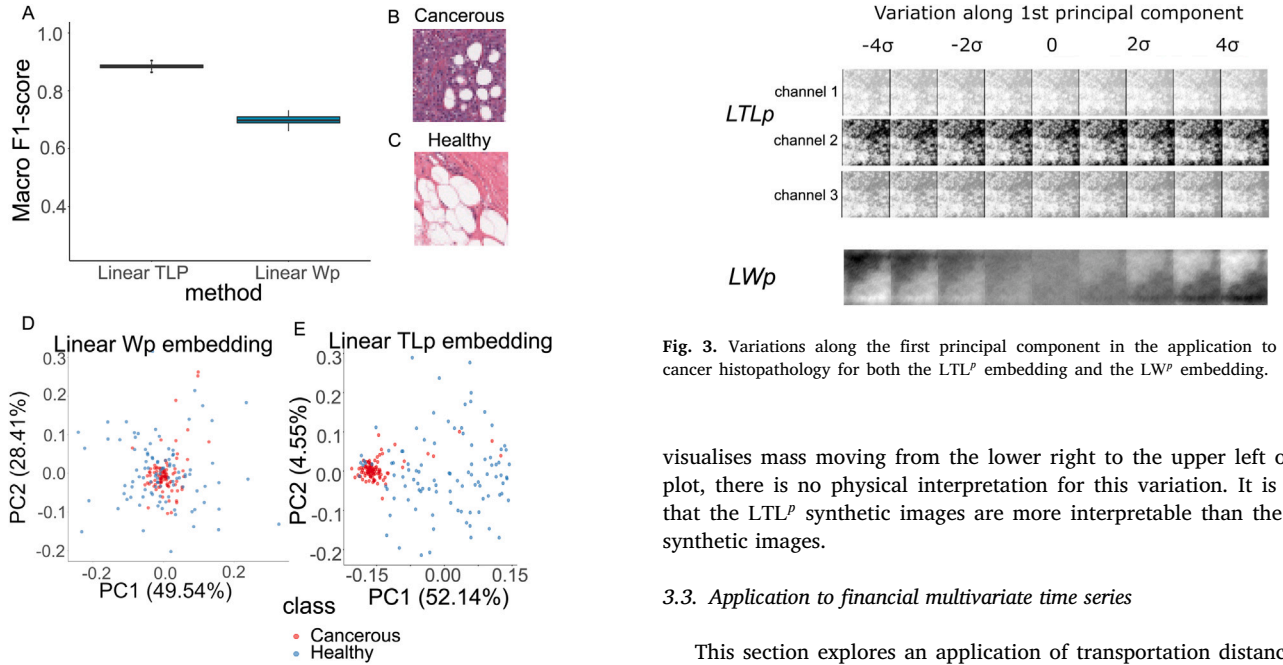


Fig. 2. (A) Distributions of macro F1 scores for LW^p and LTL^p embeddings in the application to breast cancer histopathology using the 1NN classifier. (B) An example image from a healthy patient. (C) An example image from a patient with breast cancer. (D, E) PCA plots using the LTL^p embedding and the LW^p embedding, we observe that LTL^p embedding separates classes.

cancerous and healthy images separate more obviously in the PCA representation of the LTL^p embeddings. This is supported when using the 1NN classifier, where a mean macro-F1 score of 0.70 is reported in the LW^p case, whilst for LTL^p the mean macro-F1 score is 0.88 representing a greater than 25% improvement. It is clear from the box plots that LTL^p outperforms LW^p (T-test $p < 10^{-16}$). Using linear interpolation, we visualise perturbations, in units of standard deviation, in principal component space as synthetic images (Fig. 3). The interpolation in the embedding produced by LTL^p demonstrates localised mass moving from the centre of the image towards the edges. This corresponds to cancer invading the milk ducts in cancerous tissue with open milk ducts in non-cancerous tissue. Linear interpolation in the LW^p embedding

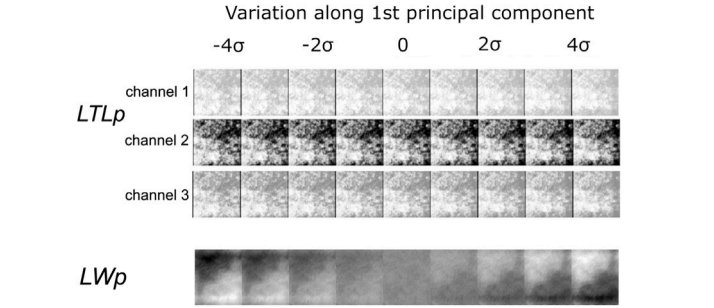


Fig. 3. Variations along the first principal component in the application to breast cancer histopathology for both the LTL^p embedding and the LW^p embedding.

visualises mass moving from the lower right to the upper left of the plot, there is no physical interpretation for this variation. It is clear that the LTL^p synthetic images are more interpretable than the LW^p synthetic images.

3.3. Application to financial multivariate time series

This section explores an application of transportation distances to financial time series. We use daily close prices for constituents of the SP1500 index, for January 2004–March 2020.

3.3.1. Experimental setup and background on financial time series

We only consider stocks available throughout the entire history, which amounts to approximately $n = 1150$ days. We use daily log-returns, with the return of stock i , between times t_1 and t_2 , defined as $R_i^{(t_1, t_2)} = \log \frac{P_{i, t_2}}{P_{i, t_1}}$, where $P_{i, t}$ denotes the price of stock i at the end of day t . On any day t , we consider the matrix $S_t = [P_{1, t-m+1}, \dots, P_{1, t}]$ of size $n \times m$ (where P_t is the vector $[P_{1, t}, \dots, P_{1150, t}]^T$), capturing the daily returns for the past m days (with fixed $m = 20$ throughout the experiments). We refer to S_t as the *sliding window*, as we vary the time component. The goal is to use and compare various techniques for computing the k -nearest-neighbors ($kNNs$) of S_t , which we denote by

$$\mathcal{N}_k(S_t) = \{t_j : j = 1, \dots, k \text{ s.t. } S_{t_1}, S_{t_2}, \dots, S_{t_k} \text{ are the } kNNs \text{ of } S_t\},$$

by pooling together the similarities between the multivariate time series comprising S_t with prior historical time series $S_m, \dots, S_{t-h}$, where h is the future horizon at which we aim to predict. Note that, at any



Fig. 4. Schematic diagram of our pipeline. We use a sliding window approach, where the current window is shown in blue, and the historical k-nearest-neighbor windows in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

given time t , the available history to query for the k-nearest-neighbors of S_t ends at S_{t-h} , in order to avoid look forward data snooping. Fig. 4 is a schematic diagram of our pipeline.

Future returns. We consider forward looking returns (referred to as targets) of two different types, and at various horizons. The future return can be the raw returns itself, or various decompositions of it. We let $f_{i,t}^{(h,RR)} = \log \frac{P_{i,t+h}}{P_{i,t}}$ denote the raw return of stock i at time t , with future horizon h . In our setting, we consider the following future horizons $h \in \{1, 3, 5, 10\}$ days. One often uses the S&P500 as a proxy for the entire market return. The S&P500 is a stock market index that measures the stock performance of 500 large companies listed on stock exchanges in the US market. The index has a corresponding ETF (Exchange Traded Fund), which can be traded much like any other regular stock (with symbol SPY), and we denote its raw return by $f_{SPY,t}^{(h,RR)} = \log \frac{P_{SPY,t+h}}{P_{SPY,t}}$. With this in mind, for any stock i , one can then consider its future *market-excess return* (MR) as $f_{i,t}^{(h,MR)} = f_{i,t}^{(h,RR)} - \beta_i f_{SPY,t}^{(h,RR)}$. For simplicity, we let $\beta_i = 1$ across all stocks $i = 1, \dots, n$, though there are various techniques to infer the individual betas from historical prices [73]. Note that the main reason for evaluating our predictions against market-excess returns, as opposed to only raw returns, is essentially to hedge away the market risk and increase the Sharpe Ratio score defined further below.

Estimates of future returns. Once we have identified the k-nearest-neighbor (knn) periods of the current window S_t , the prediction made at time t , for a given horizon h , is a weighted sum of the corresponding historical future returns, where the weights are inversely proportional to the knn distances. More precisely, if we denote by $\hat{f}_t^{(h,RR)}$ the $n \times 1$ vector of forecasts for the future h -day raw returns at time t , its values are given by

$$\hat{f}_t^{(h,RR)} = \sum_{t_j \in \mathcal{N}_k(S_t)} w_j f_{t_j}^{(h,RR)},$$

with weights given by $w_j \sim \frac{1}{d(S_t, S_{t_j})}$, normalised such that $\sum_{t_j \in \mathcal{N}_k(S_t)} w_j = 1$, where $d(S_t, S_{t_j})$ denotes the distance between the current window S_t , and its nearest historical neighbors $S_{t_j}, t_j \in \mathcal{N}_k(S_t)$. Similarly, we compute estimates for the future h -day market-excess returns (MR), by pooling together k-nn historical forward looking market-excess returns.

PnL. For performance evaluation, we rely on standard metrics from the finance literature. Given a set of forecasts, the corresponding PnL (Profit and Loss) on day t for a given return f_t (chosen to be the raw return $\hat{f}_t^{(h,RR)}$ or the market excess return $\hat{f}_t^{(h,MR)}$) is defined by

$$\text{PnL}_t = \sum_{i=1}^n \text{sign}(\alpha_{i,t}) \cdot f_{i,t}, \quad t = 1, \dots, T,$$

where $\alpha_{i,t}$ denotes our forecast for stock i on day t . Note that the PnL increases if and only if the sign of the forecast α agrees with the sign of the future return $f_{i,t}$, and decreases otherwise. The sum is across all the n stocks, and $f_{i,t}$ is the future return (either raw return (RR) or

market-excess return (MR)) of stock i on day t . We explore different forward looking horizons $h \in \{1, 3, 5, 10\}$. We add a superscript to the PnL calculation to indicate its dependency on horizon h and the type of return considered (RR or MR). For instance, in our setting, the h -day forward looking market-excess return, computed daily, is given by

$$\text{PnL}_t^{(h,MR)} = \sum_{i=1}^n \text{sign}(\hat{f}_{i,t}^{(h,MR)}) \cdot f_{i,t}^{(h,MR)}, \quad t = 1, \dots, T, \quad h \in \{1, 3, 5, 10\}.$$

Sharpe Ratio. After computing the daily PnL time series for all available days, we capture the risk-adjusted performance by computing the corresponding (annualized) Sharpe Ratio

$$\text{Sharpe Ratio (SR)} := \frac{\text{mean}(\text{PnL})}{\text{stdev}(\text{PnL})} \times \sqrt{252},$$

where the scaling is due to the fact that there are 252 trading days within a calendar year. For simplicity, we apply the same scaling $\sqrt{252}$ also to the longer horizons $h > 1$, and refer the reader to [74] for an in depth discussion on Sharpe Ratios³ and practical considerations arising from the fact that typical forecasts for equity returns are usually serially correlated. We attribute the future h -day PnL to each day t (leading to overlapping windows), as opposed to maintaining h parallel portfolios, and computing their daily total PnL. We are mainly interested in relative performance of the methods, in terms of Sharpe Ratio and PnL, and less on the actual magnitudes of these performance metrics, and their practical considerations.

Average PnL in basis points. In the financial literature, a typical performance measure is the average return per dollar traded, in percentage. For example, one is typically interested in the annualized return of the portfolio. In what follows, we denote by PPT (PnL Per Trade) the average daily PnL per unit of notional (eg., \$1). For instance, if at time t_0 the available capital is \$100, and the cumulative PnL at time t_{252} (thus after one year) is \$10, then the annualized return amounts to 10%. Recalling that 1% amounts to 100 basis points (bpts), an annualized return of 10% translates to approximately PPT = 4 bpts per day (since $4 \times 252 \approx 1000$ bpts, which amounts to 10%). Essentially, the PPT is telling us how much would we earn for each \$1 traded in the markets (excluding transaction costs and fees). For simplicity, we ignore sizing/liquidity effects and assume that each day, we invest \$1 for each of the n stocks, which leads us to the following simplified notion of PPT, averaged over the entire trading period comprised of T days

$$\text{PPT} = \sum_{t=1}^T \frac{\text{PnL}_t}{n} = \frac{\sum_{t=1}^T \text{PnL}_t}{Tn}.$$

³ The annualized Sharpe Ratio is calculated from daily observations as $\frac{\mu - r_f}{\sigma} \sqrt{252}$, where μ is the average daily PnL return, r_f denotes the risk-free rate, and σ the standard deviation of the PnL returns. Since the risk-free rate is close to zero over the period of study, we compute the Sharpe Ratio as $\frac{\mu}{\sigma} \cdot \sqrt{252}$.

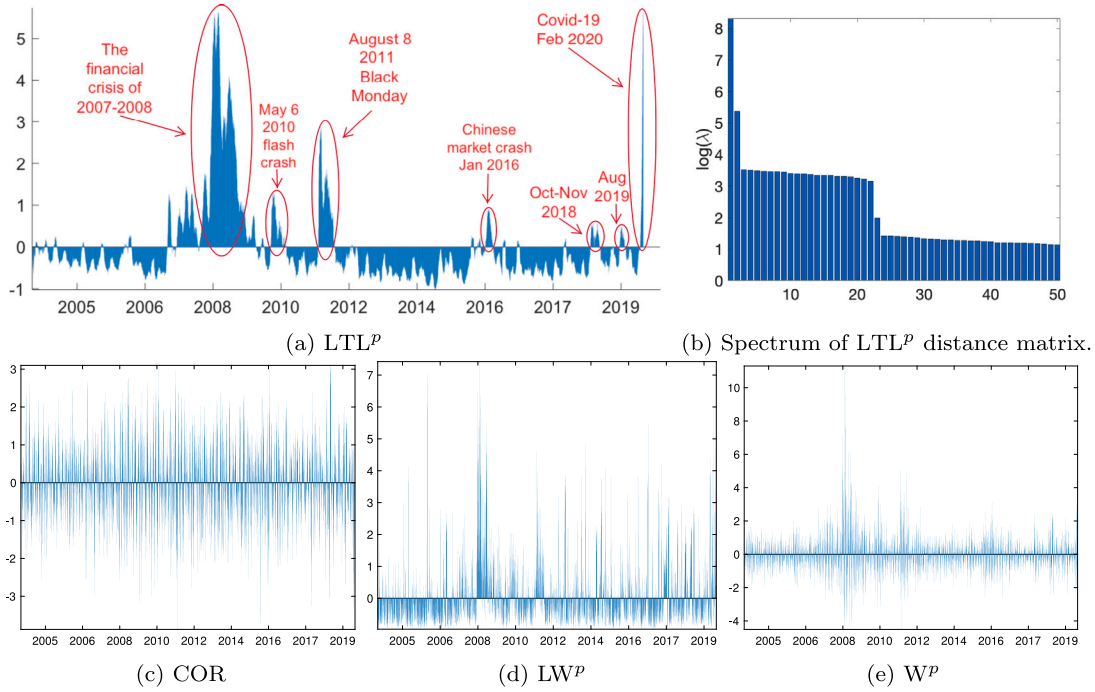


Fig. 5. Normalized total distance for each day, as computed by LTL^p (a), COR (c), LW^p (d) and W^p (e). We annotated plot (a) LTL^p with the major market events. More explicitly, we compute the distance matrix between all pairs of days (where the data for a given day is given by the previous $m = 20$ days, including the day of), construe this as a distance network, and compute the total degree of each node (which we show in the above figures (a,c,d,e), after standardisation, for ease of visualization). Plot (b) shows a barplot of the top eigenvalues of the LTL^p pairwise distance matrix.

Quantile portfolios. One is often interested in understanding the performance of the forecasts, as a function of their respective magnitudes. To this end, one typically considers only a subset (eg. top $q\%$ strongest in magnitude forecasts) of the universe of stocks, usually referred to as *quantile portfolios* in the literature [75]. A quantile-based analysis simply constructs and evaluates portfolios composed of stocks which fall in a specific quantile bucket, or above a quantile threshold. We choose to use upward-contained quantile buckets, which we denote by $qr_i, i = 1, 2, \dots, 5$ indicating the quantile rank of each stocks, meaning that stocks with quantile rank qr_i correspond to the top $1 - \frac{i-1}{5}$ fraction of largest-in-magnitude forecasts. The colours in Figs. 7 denote the quantile portfolios traded based on the magnitude of the forecasts. For example, the red bars qr_1 correspond to the full universe of stocks, while the green bars qr_4 denote the top 40% largest in magnitude forecasts.

3.3.2. Methods comparison

We compare the prediction performance of W^p , LW^p and LTL^p , and leave out the full TL^p due to its prohibitive computational running cost. In addition, we compare to another more classical approach, not relying on transportation distance methodology, given by the simple Pearson correlation between the original time series. More precisely, for each window S_t (matrix of size $n \times m$, $t = 1, \dots, T$), we first standardise the returns in each row (eg. corresponding to each stock), and denote the resulting matrix by $\tilde{S}_t, t = 1, \dots, T$. Next, we unwrap each matrix $\tilde{S}_t$ into a vector $\psi_t \in \mathbb{R}^{nm}$, and finally compute the pairwise distance between a pair of time windows S_i and S_j , using a correlation-based distance between their corresponding flattened versions: $COR_{ij} := 1 - \text{Corr}(\psi_i, \psi_j)$. Note that, in light of the pre-processing step that standardised each stock, this corresponds, up to a scaling constant, to the squared Euclidean distance between the corresponding vectors $\|\psi_i - \psi_j\|_F^2$.

Fig. 6 shows numerical results comparing the various methods considered. The left column is a heatmap showing the $T \times T$ pairwise distance matrix between all days available in history, in the interval 2004–2020. We note that LTL^p clearly highlights the financial crisis occurred in 2008, followed by LW^p , and to some extent, W^p , while COR show barely visible signs of this event. The middle columns show a distribution of the pairwise distances, while the right columns show the row sums of the distance matrix. Construing the distance matrix as a network with distance/dissimilarity information, this plot effectively plots the degree of each node (i.e., of each time period corresponding to a sliding window of length 20). The periods of time with the largest dissimilarity degree correspond to the financial crisis in 2008. Note that, for each of visualisation, we standardise the degree vector.

Next, we zoom in on the LTL^p pairwise distance matrix, and show the resulting degrees in Fig. 5. After computing the LTL^p distance matrix, we interpret it as a distance network, and compute the total distance degree of each node, after standardisation. In this plot, we are able to recognise many of the major financial market events occurred over the last two decades: the big financial crisis of 2007–2008, the 2010 Flash crash, the August 2011 markets fall (between May–October 2011), the Chinese market crash from January 2016, the period Oct–Nov 2018 (when the stock market lost more than \$2 trillion), August 2019 (a highly volatile month in the global stock markets), and finally, the February 2020 stock market crash triggered by the COVID-19 pandemic. It is interesting to observe that the distance degree corresponding to the COVID-19 pandemic is matched in magnitude only by the 2007–2008 financial crisis. Furthermore, the top right plots of Fig. 5 show the top 50 eigenvalues of the LTL^p distance matrix, highlighting the usual market mode top eigenvector, with the second eigenvector highly localised on the 2007–2008 financial crisis and the Feb–March 2020 Covid-19 pandemic period (the top 5 eigenvectors are shown in the Appendix, see Figure B.13).

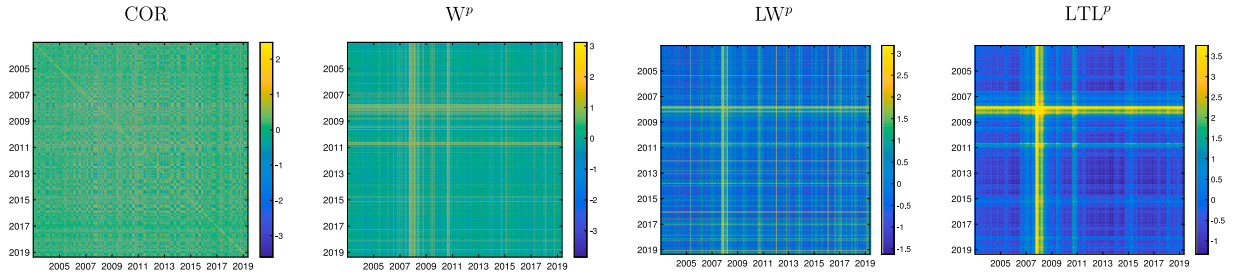


Fig. 6. Pairwise distance matrices corresponding to each of the methods considered.

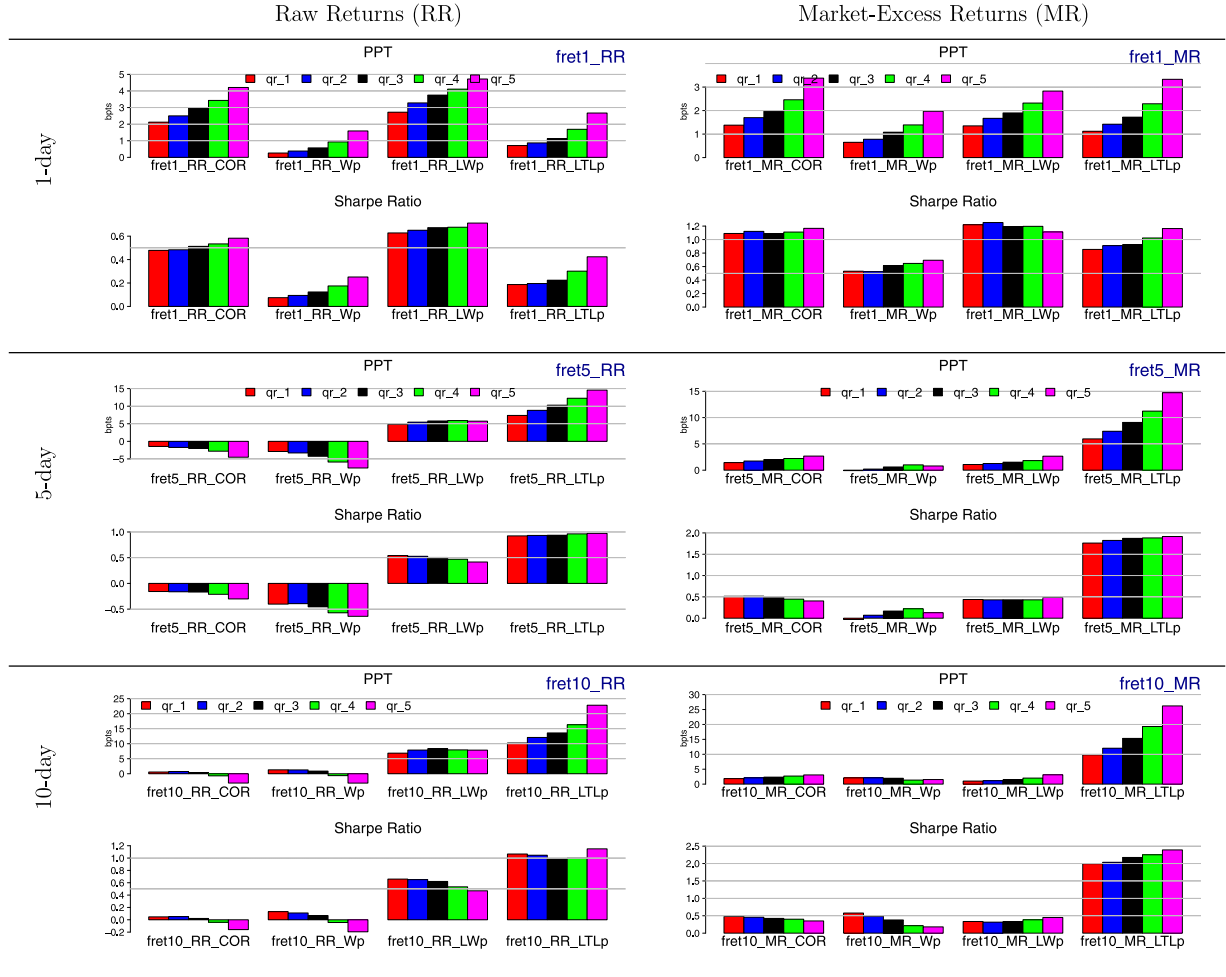


Fig. 7. Portfolio statistics across various future horizons $h \in \{1, 5, 10\}$ across two types of returns: raw returns and market-excess returns. The colours denote quintile portfolios, and the x-axis denotes the forecasts made by each method $\{COR, W^p, LW^p, LTL^p\}$.

Fig. 7 shows portfolio statistics of the methods, across different target future horizons $h \in \{1, 5, 10\}$, for both raw-returns (RR) and market-excess returns (MR). The corresponding cumulative PnL plots across time are given in Fig. 8 for various future horizons. We fix the number of nearest neighbors to $k = 100$, and allow the knn search to span back until the start of the available history $T = 1$. When forecasting raw returns (left column in Fig. 7), all methods perform rather poorly, with COR and LW^p showing the best performance for $h = 1$, while for $h \in \{5, 10\}$, LTL^p clearly outperforms all other methods. This supports the assumption that TL^p is better able to model similarities in financial time series.

For market-excess returns, when $h = 1$, all methods return a similar performance in terms of Sharpe Ratio (SR) around 1, except for W^p which has a SR of around 0.5; in terms of PnL, most methods achieve

a PPT of 1–3 basis points (bpts). However, for longer horizons, LTL^p clearly outperforms all other methods, both in terms of Sharpe Ratio and PPT.

4. Discussion

We have presented a linear optimal transport distance for use in pattern recognition tasks based on the TL^p distance. The TL^p distance generalises the Wasserstein distance such that it can be applied to multi-channelled data, for example colour images and multivariate time-series. However, when pairwise distances are needed the computation of the TL^p distance can make its routine application in pattern recognition tasks computationally infeasible. We proposed a method to alleviate this problem by extending the LW^p framework to the TL^p

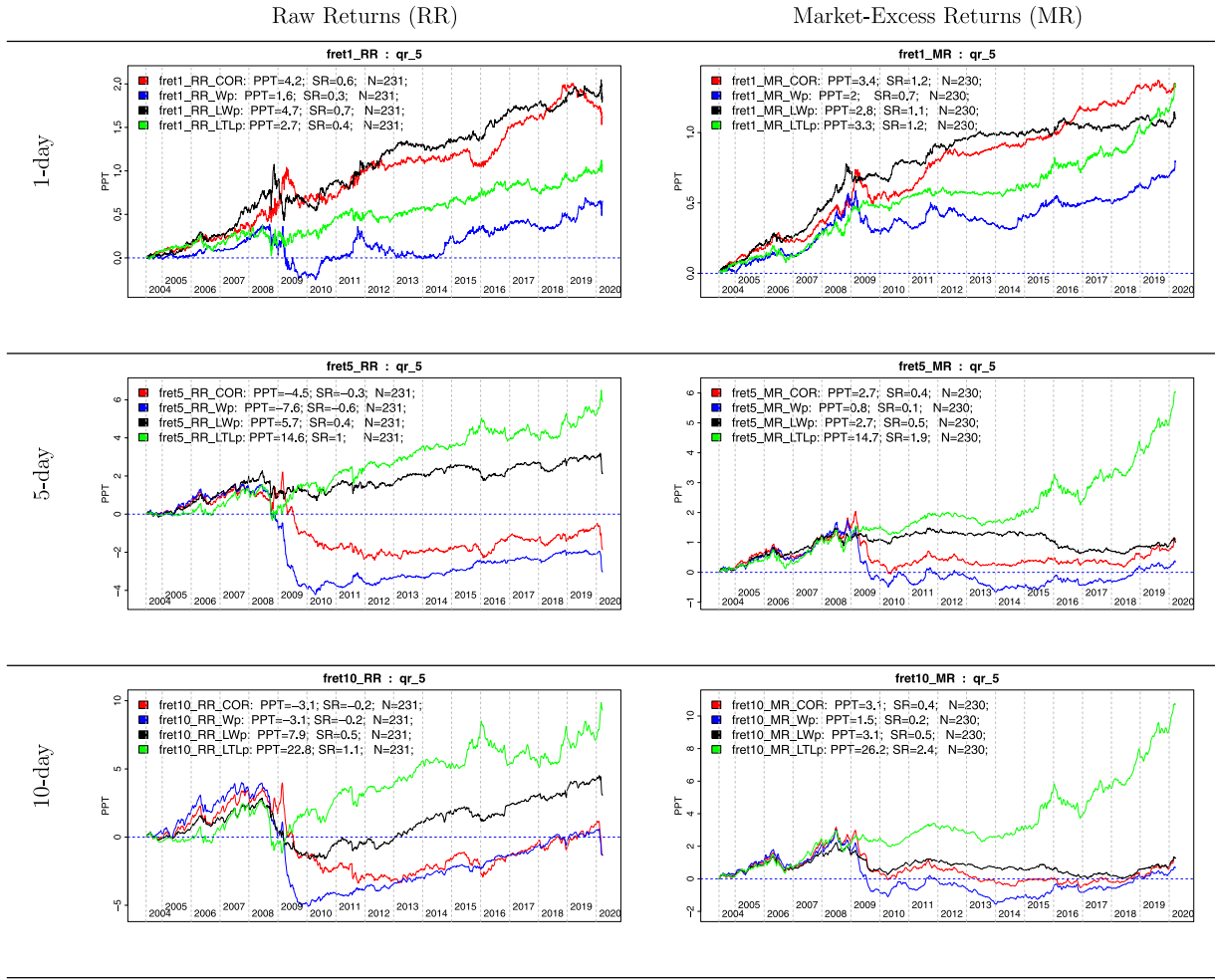


Fig. 8. Cumulative PnL of the top quintile portfolio qr_5 for the 1,5,10-day future horizons, for raw returns and market-excess returns, across different methods. The legend contains performance statistics of each forecast method: PnL Per Trade (PPT) in basis points, Sharpe Ratio (SR), while N denotes the size of the portfolio (since qr_5 contains the top 20% largest magnitude forecasts, this amount to $N \approx 1150/5 = 230$ stocks).

setting and call this the LTL^p framework. In a dataset of N signals, these linear transportation approaches need to only compute N transport maps/distances and thus the cost scales linearly in the number of signals. In contrast, if pairwise distance are needed, the cost scales quadratically with the number of signals, which can render the problem infeasible.

We developed the theory required to allow these transportation methods to be applied. Showing the preservation of these transportation distances with respect to a reference signal, as well as showing that the LTL^p defines a bona fide metric on the TL^p space. We additionally showed that the linear transportation methods facilitates a linear embedding of our signals allowing simple statistical methods to be applied to the data.

We compared our methods on Auslan, Breast Cancer Histopathology and Financial times series problems. In each case, the LTL^p approach significantly outperformed the LW^p approach. Furthermore, this came at minimal increased computational cost. Even though TL^p often outperformed LTL^p on this task this improved performance comes at an unreasonable additional computational cost.

Our method still relies on the computation of transport maps and this comes at a cost. We have found entropy regularised methods to perform well but suffer from instability. Fast and stable algorithms which are memory efficient are still needed by the community for reliable computation of transport maps.

Extensions of the TL^p framework could be to the manifold setting [76], which would allow computation of the TL^p distance on a

graph. The TL^p barycentre and other transformations have also yet to be explored. Considering TL^p has a spatial penalty it could also be combined with other penalty terms for more complex applications, for example one could also include a penalty on derivatives of signals as in the $TW^{k,p}$ distance, see [43].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used in this manuscript is open-access.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2023.110080>.

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Further reading

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